

*The 2024 Taiwan International Mission Idea Contest on CubeSat
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Taiwan-India Lunar Mapping Constellation (TILMaC)

NIKHIL RIYAZ, INBISAT YOUSUF, SHUVAM PAL, SAINATH, ASHWIN REDDY

DR. K GOPALAKRISHNAN, DR. S THANGAVELU

Sri Shakthi Institute of Engineering & Technology

Coimbatore, Tamil Nadu, India

nikhilryz@gmail.com, inbisatyoufufnath@gmail.com

THIRU VENKAT R, SYED HAYATH, MD. WAEL UMAR, ROSHANI BANKAR, RAMYA SHREE R

Cambridge Institute of Technology

KR Puram, Bangalore, India

thiruveenkat.er@gmail.com, waelraed4281@gmail.com

PROF. YING LIAO

National Taipei University of Technology

Zhongxiao E. Rd., Taipei, Taiwan

yliao@ntut.edu.tw

ABSTRACT

The Taiwan-India Lunar Mapping Constellation (TILMaC) mission is a pioneering initiative deploying four 6U CubeSats to conduct high-resolution mapping of the lunar surface and subsurface. This mission leverages Dual-Frequency Synthetic Aperture Radar (DFSAR) and Hyperspectral Imaging (HSI) to achieve key objectives: water-ice detection, mineral composition analysis, monitoring regolith change and mapping magnetic anomalies. Positioned in a circular lunar orbit, TILMaC captures comprehensive spatial and temporal data supporting future exploration and resource utilization on the moon.

This paper details TILMaC's mission objectives, constellation architecture and payload design including specific capabilities of the DFSAR and HSI systems for advanced surface analysis. It also discusses the CubeSat configuration, data relay strategy providing insights into subsystem design, risk management and data processing.

KEYWORDS: Moon, Hyperspectral Imaging, Mapping, Constellation, SAR

1. INTRODUCTION

The proposed satellite constellation consists of four satellites, two equipped with Dual-Frequency Synthetic Aperture Radar (DFSAR) and two with Hyperspectral Imaging (HSI) capabilities. These satellites will be positioned in a circular orbit with 90-deg inclination maintaining an altitude of 100 km above the lunar surface. The constellation design features the satellites pointing inwards with one satellite in each pair-oriented forward along the orbital path and the other oriented backward allowing the satellites to capture images from different slant angles facilitating a 3D construction of the lunar surface. DFSAR satellites will generate high-resolution topographic maps while HSI satellites will provide detailed compositional data. Combined, both technologies will enable geological mapping, resource identification and analysis of lunar surface construct and

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composition providing insights for detailed investigations of lunar geology.

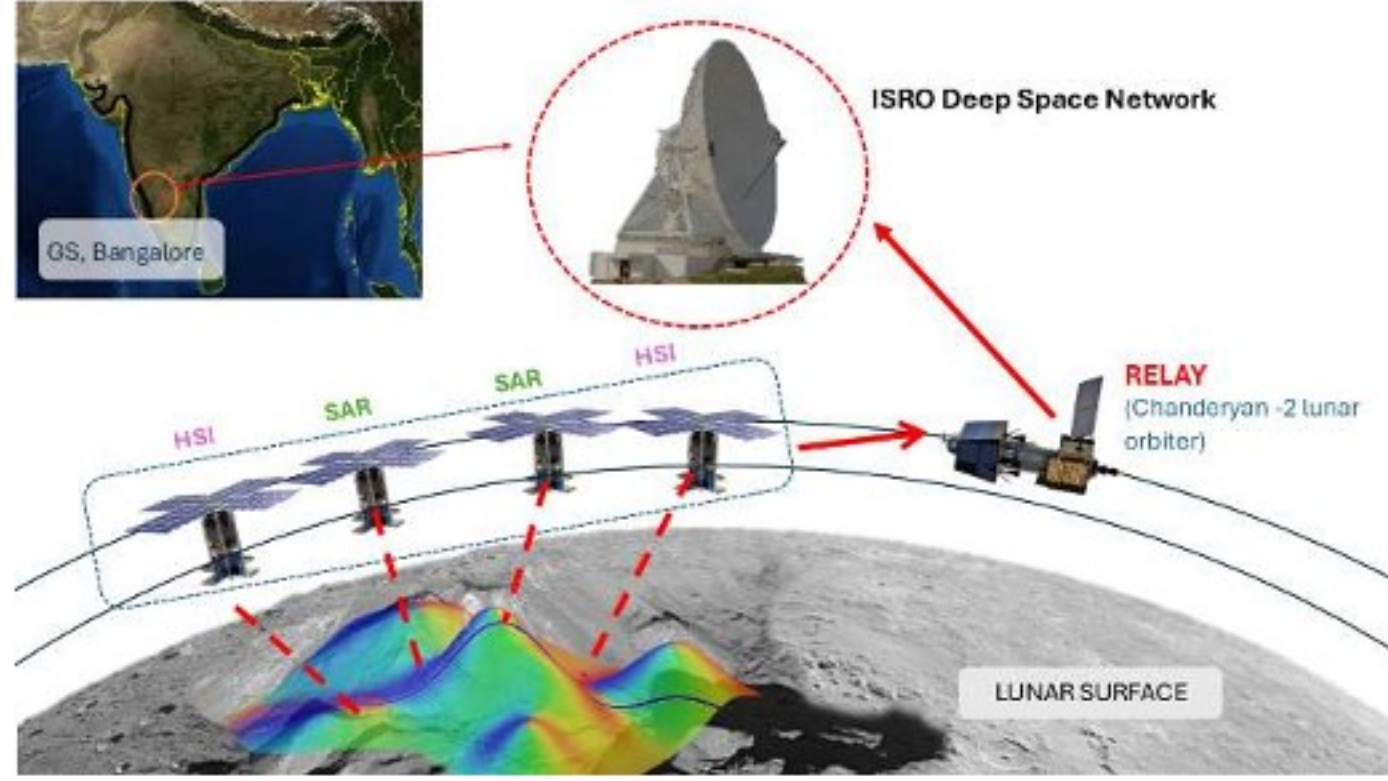


Figure 1: Concept of Operations

2. MISSION OBJECTIVES

2.1 Lunar Surface Mapping: Utilizes Synthetic Aperture Radar (SAR) to map the lunar surface with a spatial resolution of 3-10 meters.

2.2 Water-Ice and Mineral Detection: Detect the presence of water-ice and analyze the composition of valuable minerals on and beneath the lunar regolith by combining (HSI) and SAR technologies.

2.3 Regolith Change Detection: Identify changes in the lunar regolith with HSI and SAR, at resolutions of 3 to 75 meters to assess alterations caused by seismic activity due to meteorite impacts or human activities.

2.4 Lunar Magnetic Field Mapping: Employ HSI to detect and map minerals combined with DFSAR to map surface and subsurface magnetic anomalies. This objective provides insights into the moon's magnetic history and aids in locating regions with concentrated magnetic materials.

2.5 Surface Composition Monitoring: Identify and classify various minerals based on spectral signatures using HSI at 400–1700 nm with a resolution of 5–10 meters. Also analysing the effects of the lunar atmosphere and micrometeorite

impacts on surface mineral compositions over time.

2.6 Lunar Dust and Gas Monitoring: Capture fine particle movements and surface changes using multiple sensors. HSI and Interferometric SAR (InSAR) will provide temporal analysis, while machine learning algorithms study dust and gas dynamics.

3. PAYLOADS

3.1 Hyperspectral Imager:

The optical system for HSI features a monolithic refractor telescope with a 5 cm aperture and a field of view (FOV) adjusted to achieve a 5-meter ground sample distance (GSD) at an altitude of 100 km corresponding to a FOV of ~2.86 degrees. The spectrometer is optimized for 400-1700 nm with a 25 μm slit and high-quality achromatic collimating lens. The housing includes damping mounts providing vibration isolation and thermal control including multi-layer insulation blankets and heat radiators.

Sensor type	Line scan
Spectrum Capability	SWIR (950-1700 nm)
Sensor Technology	CMOS

Table 1: Sensor 1 Configuration

Sensor type	Line Scan
Spectrum Capability	Visible (400-700 nm), Near Infrared (700-1000 nm)
Sensor Technology	CMOS

Table 2: Sensor 2 Configuration

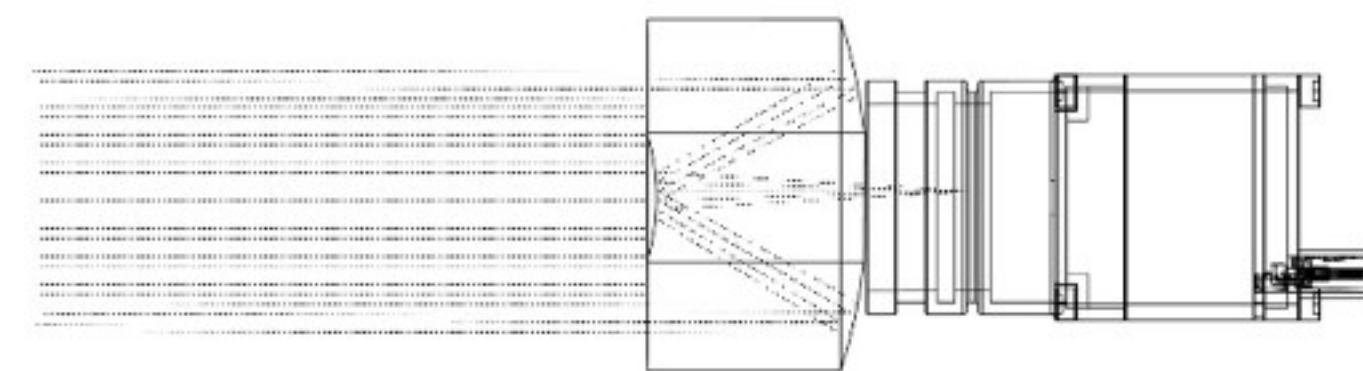


Figure 2: HSI Ray Tracing

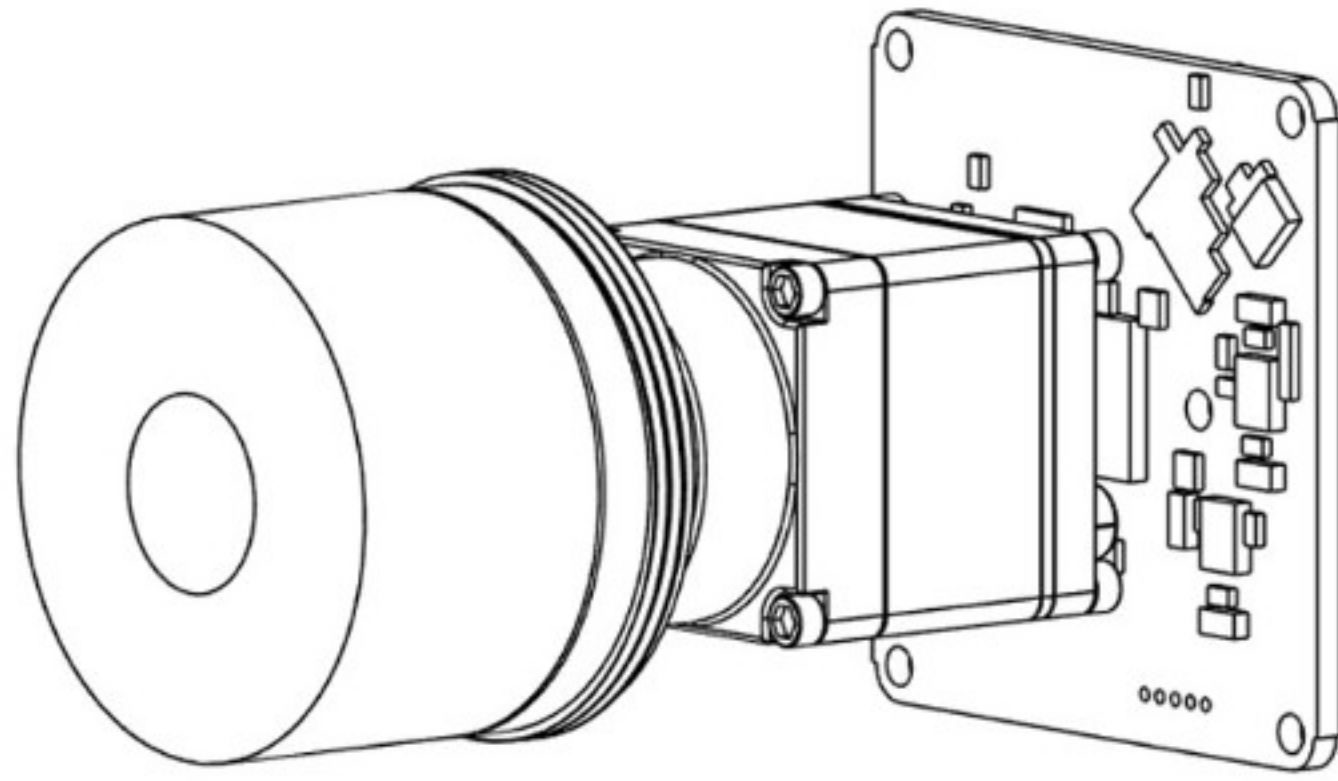


Figure 3: HSI Module

3.1.1 Overview of Hyperspectral Imaging

Applications: HSI helps differentiate minerals based on unique absorption features within specific wavelength ranges. For lunar mineral mapping, HSI is particularly effective in the visible to near-infrared (VNIR) range (0.42–3.0 μm), which contains diagnostic spectral signatures for critical minerals such as olivine, pyroxene, and plagioclase. The Moon Mineralogy Mapper (M3) on Chandrayaan-1 demonstrated HSI's ability to map mineral composition and detect water-ice deposits. (Pieters et al., 2009)

3.1.2 Detection of Trace Minerals and Specific Bands

Bands: Iron-bearing minerals like pyroxene and olivine display strong absorption features at ~ 1000 nm and ~ 2000 nm respectively. These wavelengths correspond to electronic transitions in iron allowing precise identification and spatial distribution mapping of iron-bearing minerals on the lunar surface. (Pieters et al., 2009)

3.1.3 Identification of Iron and Detection of Magnetic Fields in Moon Swirls

Iron-rich minerals are often found in regions with unique surface features such as moon swirls, which are hypothesized to be influenced by magnetic fields. These swirls may contain ferromagnetic materials detectable by HSI through their unique spectral signatures in the VNIR range. By identifying the presence and concentration of ferromagnetic minerals within these areas, HSI can offer insights into whether magnetic fields contribute to swirl

formation. (Green et al., 2011; Zuo et al., 2024).



Figure 4: Swirls East of Firsov Crater, from Apollo 10

3.1.4 Proposed Hyperspectral Imager and Methodology

Methodology: The HSI payload is designed to operate across 400–1700nm, providing a GSD of ~ 5 meters. This spectral range includes key bands for detecting silicates, oxides, and hydroxyl-bearing minerals. The imager employs a monolithic refractor with a 5 cm aperture, capturing spectral data over a 10 km swath. Real-time data processing is achieved via an onboard FPGA with lossy/lossless compression ensuring efficient storage and transmission. This setup allows for continuous monitoring, facilitating the detection of dynamic changes in mineral compositions over time particularly in high-interest regions.

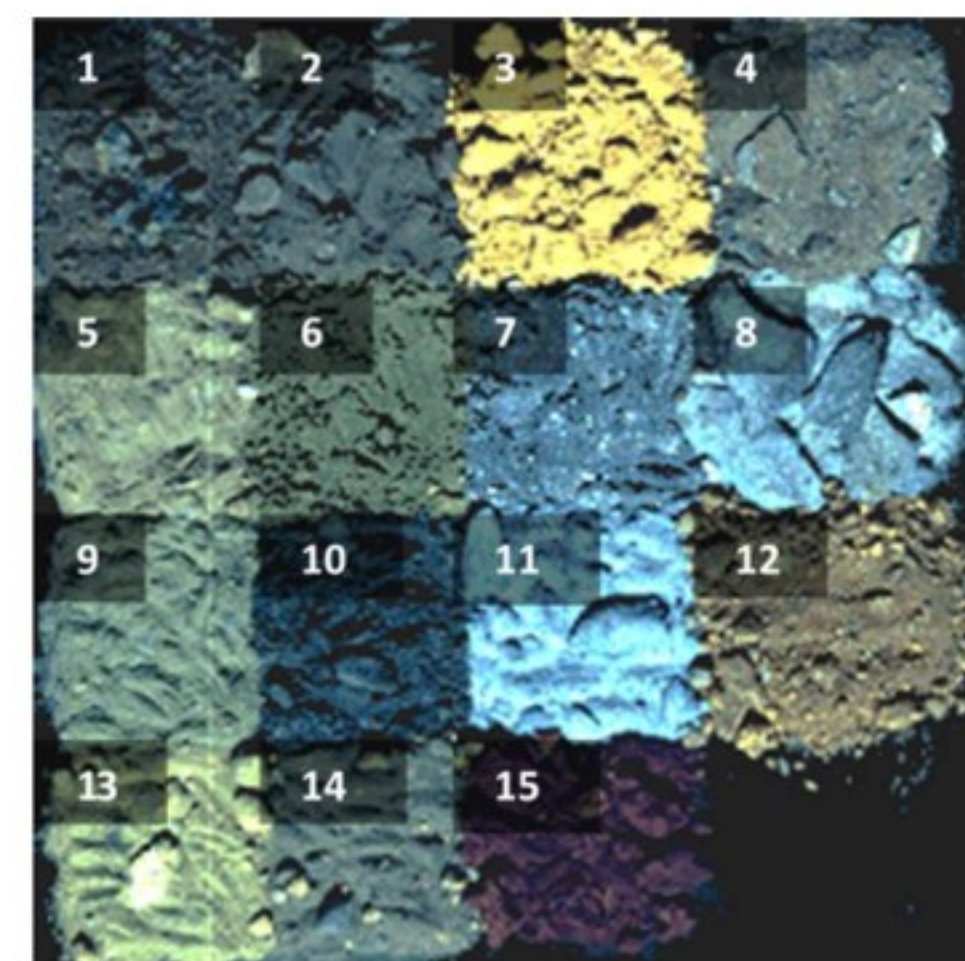


Figure 5: Iron Ore Samples

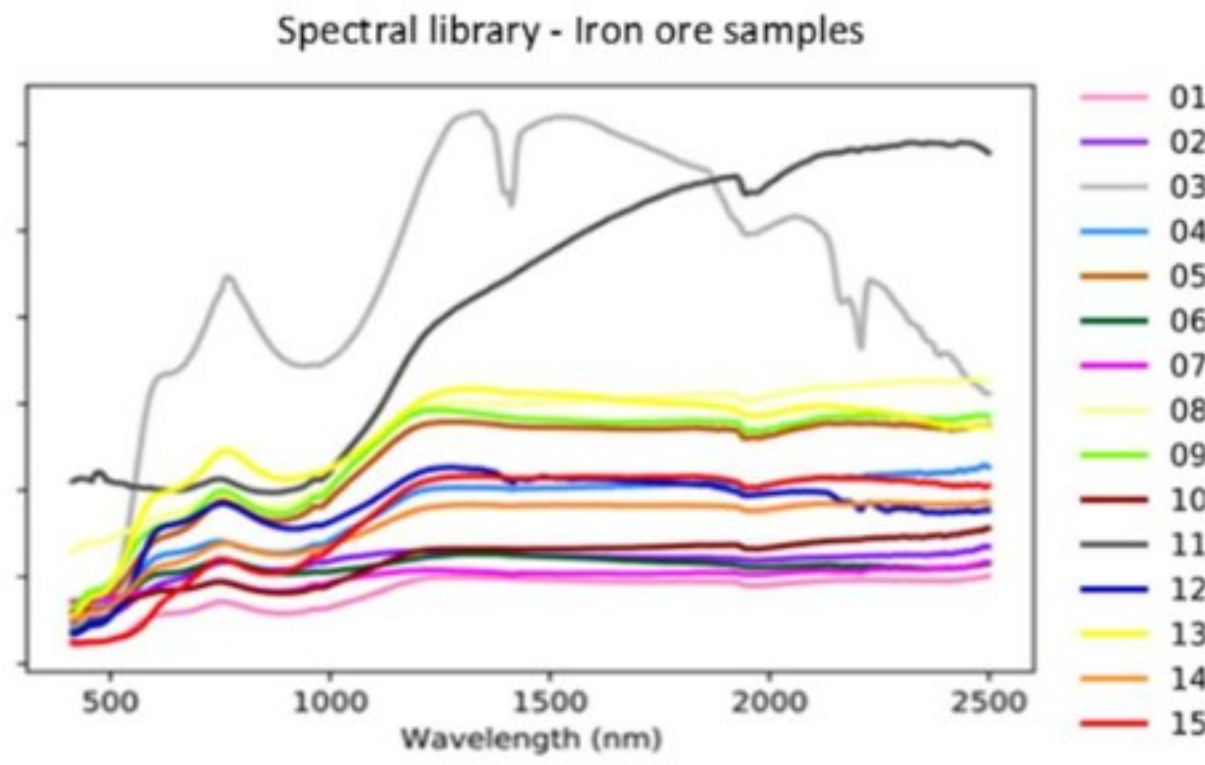


Figure 6: Spectral Absorption of Iron Ore Samples

3.1.5 Data Generation and Processing: A 10-bit analog-to-digital converter (ADC) and a digital signal processor (DSP) built around a FPGA for real-time data processing is included for electronics and data handling. Data storage comprises of a 2 TB Rad-hard SSD with interfaces like LVDS and SpaceWire and uses CCSDS 122.0-B-2 Lossy/Lossless compression. Based on the GSD, 21,84,200 frames are captured per orbit (at 140 minutes orbital period, 260 frames are captured per second). The lowest data rate at 30 bands, on a 2000x1 sensor, with a data depth of 8 bits, is 15.6 Mbps, and the maximum data rate at 148 bands with 10-bit depth is 96.2 Mbps which results in 790 GB of total data generated over an orbit before compression.

3.2 Dual Frequency Synthetic Aperture Radar (DFSAR):

The DFSAR instrument is designed to operate in dual-frequency modes, utilizing both L-band and S-band radar systems and incorporates capabilities for interferometric SAR (InSAR) and polarimetric SAR (PolSAR) imaging, allowing for analysis of scattering properties of the lunar surface across multiple look-angle geometries (Bhiravarasu et al., 2021). The DFSAR setup includes independent L-band and S-band systems sharing a common microstrip antenna enabling synchronous dual-band operation where timing and synchronization

are managed by signals from the L-band system (Putrevu et al., 2016).

3.2.1 Signal Transmission and Reception

Mechanisms: DFSAR transmits Linear Frequency Modulated (LFM), or chirp signals across a selectable bandwidth from 60 MHz to 2 MHz, using Active Electronically Steerable Array (AESA) antennas for flexible polarimetric measurements across multiple slant angles. This flexibility enables full-polarimetric imaging, capturing backscatter in HH, HV, VH, and VV polarizations which are essential for analyzing dielectric properties. (Bhiravarasu et al., 2021). The AESA system employs an origami-style deployment, using Nitinol strips to maximize surface area, enhancing imaging flexibility across various incidence angles

3.2.2 Dual-Satellite Operation and Synchronization:

Following the dual-satellite configuration like the TanDEM-X mission, TILMaC's DFSAR captures InSAR data with two satellites operating in synchronous L/S-band configurations. This allows for multi-angle phase data collection where one satellite transmits while both satellites receive. By synchronizing transmissions and varying incidence angles, TILMaC creates an interferometric baseline which can be used for generating accurate Digital Elevation Models (DEMs) like those generated in terrestrial TanDEM-X applications (Zink et al., 2006)

3.2.3 Interferometric and Polarimetric Capabilities:

Interferometric SAR (InSAR) uses a dual-satellite approach, TILMaC's DFSAR system captures interferometric data, which facilitates phase-difference analysis for DEM construction. This setup enables precise topographic mapping, allowing detailed DEM generation critical for

analyzing elevation changes and depth variations in the lunar surface (Bhiravarasu et al., 2021).

Polarimetric SAR (PolSAR) capabilities enable the capture of polarization-based scattering, providing insights into dielectric properties which are vital for differentiating surface materials and characterizing geological features. Fully polarimetric data improve mineral and regolith texture mapping offering a more nuanced understanding than single-polarization SAR (Putrevu et al., 2016)

3.2.4 Penetration Depth and Ground Resolution: DFSAR's dual-frequency design leverages the different penetration capabilities of L-band and S-band frequencies:

L-band (1.25 GHz): Penetrates up to 5 meters into the lunar regolith, allowing subsurface exploration ideal for locating buried rock layers and assessing dielectric properties associated with minerals like ilmenite.

S-band (2.5 GHz): Offers detailed surface mapping with a shallower penetration depth of 3 meters, ideal for capturing surface roughness and identifying smaller-scale geological features (Bhiravarasu et al., 2021).

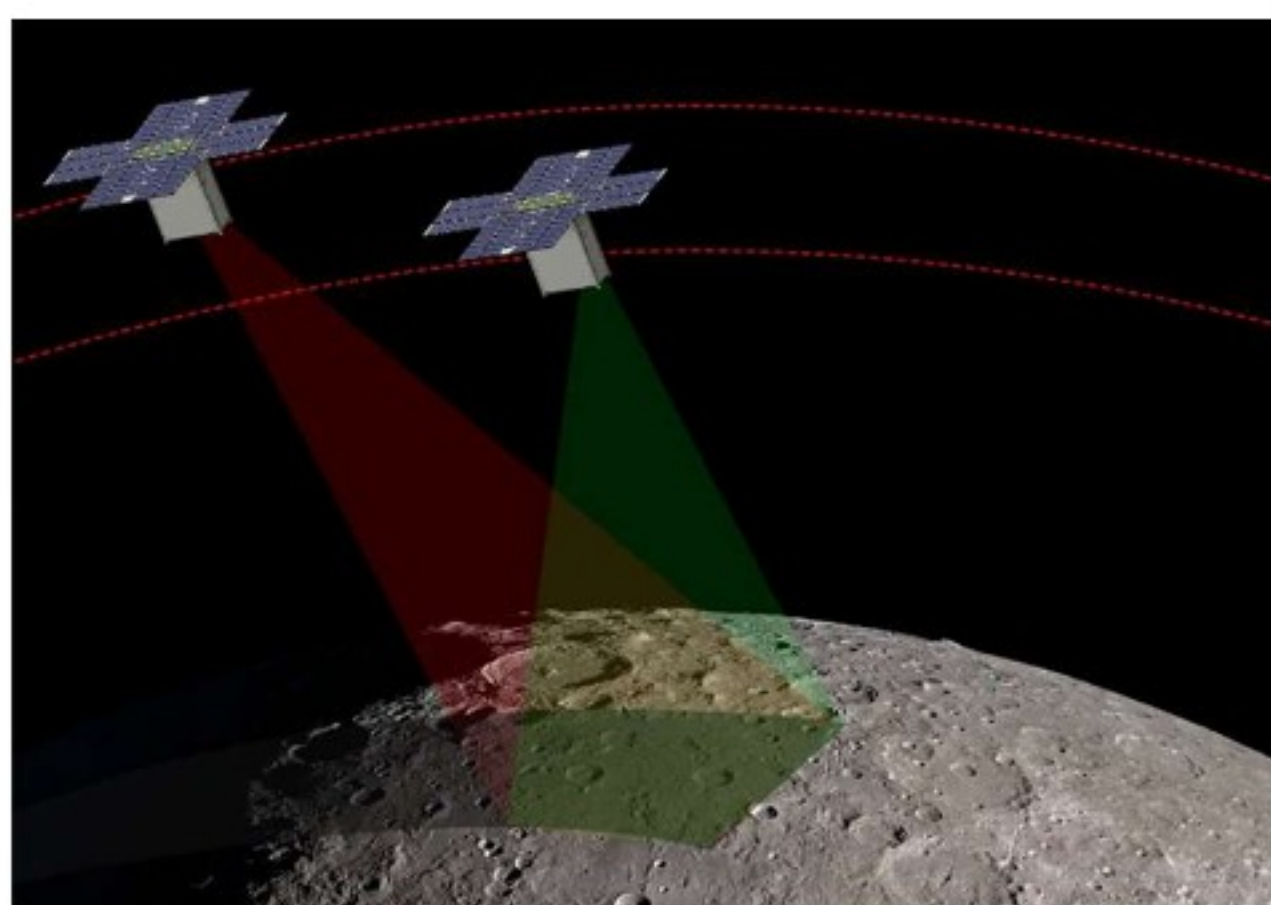


Figure 7: Satellite SAR Representation

3.2.5 Integration of DFSAR Data with Lunar Orbiter Laser Altimeter (LOLA) Data: Combining DFSAR with LOLA altimetry enhances DEM quality by enabling radar

distortion corrections and adding depth information in complex terrains such as crater walls. This synergy allows for accurate, high-resolution 3D models, leveraging LOLA's altitude data with DFSAR's InSAR-derived surface elevation for precise topographic and volumetric mapping (Liu and Jin, 2023)

3.2.6. Advantages of DFSAR over Traditional SAR: The DFSAR's dual-frequency, polarimetric, and interferometric capabilities offer significant advantages:

Improved DEM Accuracy: Through InSAR data, TILMaC produces precise DEMs, achieving greater topographical detail compared to single-band SAR, similar to TanDEM-X's terrestrial DEM applications.

Enhanced Surface and Subsurface Analysis: Dual-frequency imaging (InSAR & PolSar) and LOLA integration allow for deeper and more detailed geological analyses across both surface and subsurface layers.

Advanced Mineral Mapping and Dielectric Property Detection: Polarimetric data enhances DFSAR's ability to detect dielectric properties, enabling identification of potential water-ice and other resources within regolith, as well as studying ferromagnetic material distributions in lunar anomalies (Bhiravarasu et al., 2021)

4. ORBIT CONFIG. & COVERAGE

Altitude (h)	100 km
Semi-major axis (a)	1837.4 km
Inclination (i)	90°
RAAN	270°
	160°

Table 3: Orbital Parameters

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We chose a frozen lunar orbit, around 100 kilometers above the moon, for our mission. Keplerian orbital parameters are shown in Table 3. The semi-major axis (a) is about 1837.4 km, which combines the orbital height of 100 km with the Moon's radius of 1737.4 km. The inclination (i) is about 90° , creating a polar orbit that permits thorough lunar surface survey, while the eccentricity (e) is close to zero, guaranteeing a stable circular trajectory. We chose a Right Ascension of Ascending Node (RAAN) of 270° and ω of 160° to align the orbit for reliable imaging and data collecting to maximize the timing of overpasses for focused mapping. The value of true anomaly changes during the mission and is set to maintain ISL. Based on the stability of missions like Chandrayaan-2 and Lunar Prospector, this 100 km circular orbit is known to be stable and requires little delta-v to maintain station-keeping.



Figure 8: Orbital Simulation

Based on the Lunar Prospector's gravity measurement data, we have calculated a daily drift margin of about 0.6 km from a starting point of 100 km. This buffer is intended to account for possible variations brought on by gravity perturbations on the moon, guaranteeing reliable mission performance. (Lozier et al., 1998). Using a two-burn Hohmann transfer, orbital corrections are achieved by first converting to an elliptical orbit and then circularizing it back to 100 km. We intend to make these modifications conservatively every 50 days to lower operating risks, resulting in an

annual delta-v demand of 87.5 m/s. Each maneuver requires a delta-v of 0.25 m/s. The goal of these operations is to keep altitude changes within ± 20 km throughout each 50-day cycle, while also controlling eccentricity and argument of periapsis.

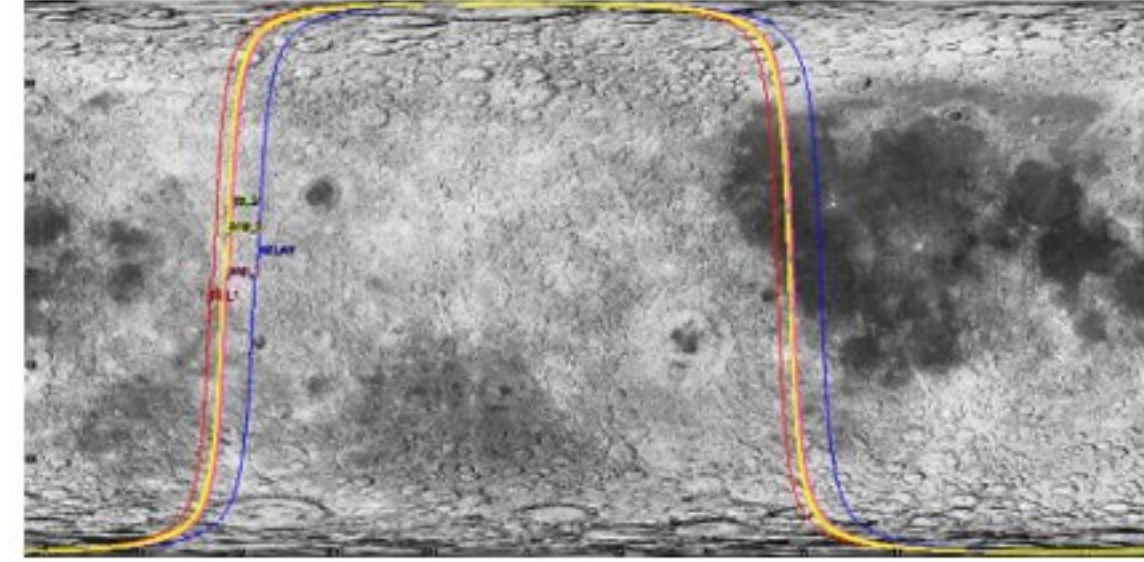


Figure 9: Trajectory Mapping of Satellites
Considering daylight constraints, and a 20% buffer for operational overlap and contingencies, we calculate that 259 days are required to guarantee full lunar surface coverage given a 10 km sweep width. As a result, 260 to 300 days is the estimated mission duration.

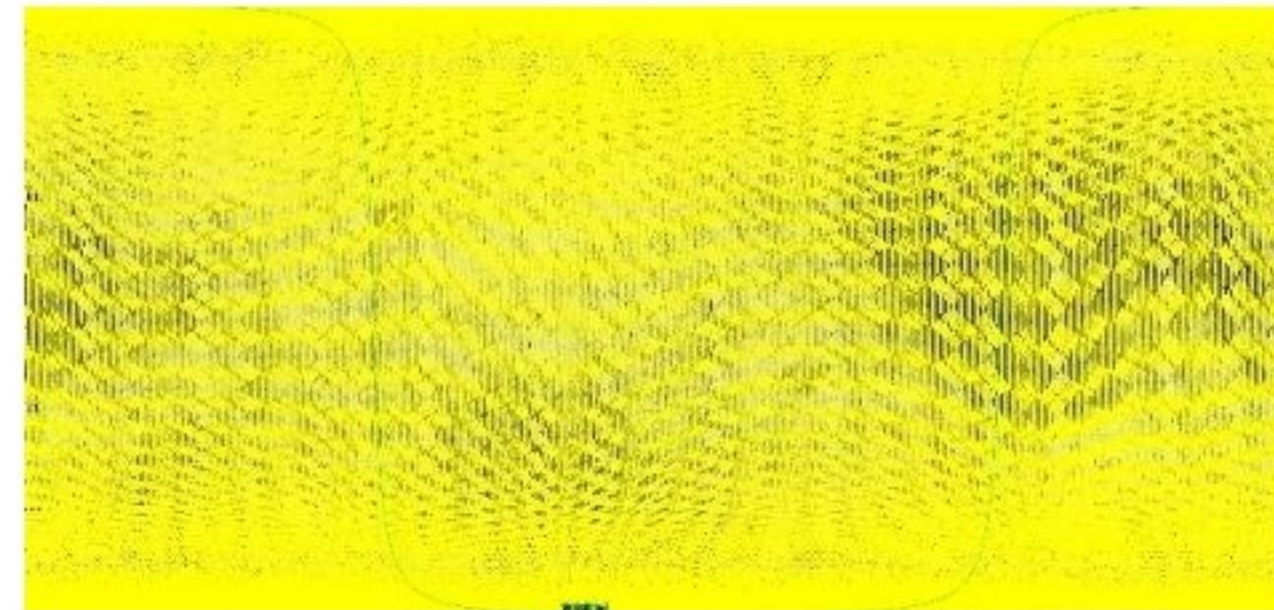


Figure 10: 100-Day Coverage @10km Swath

5. SATELLITE BUS

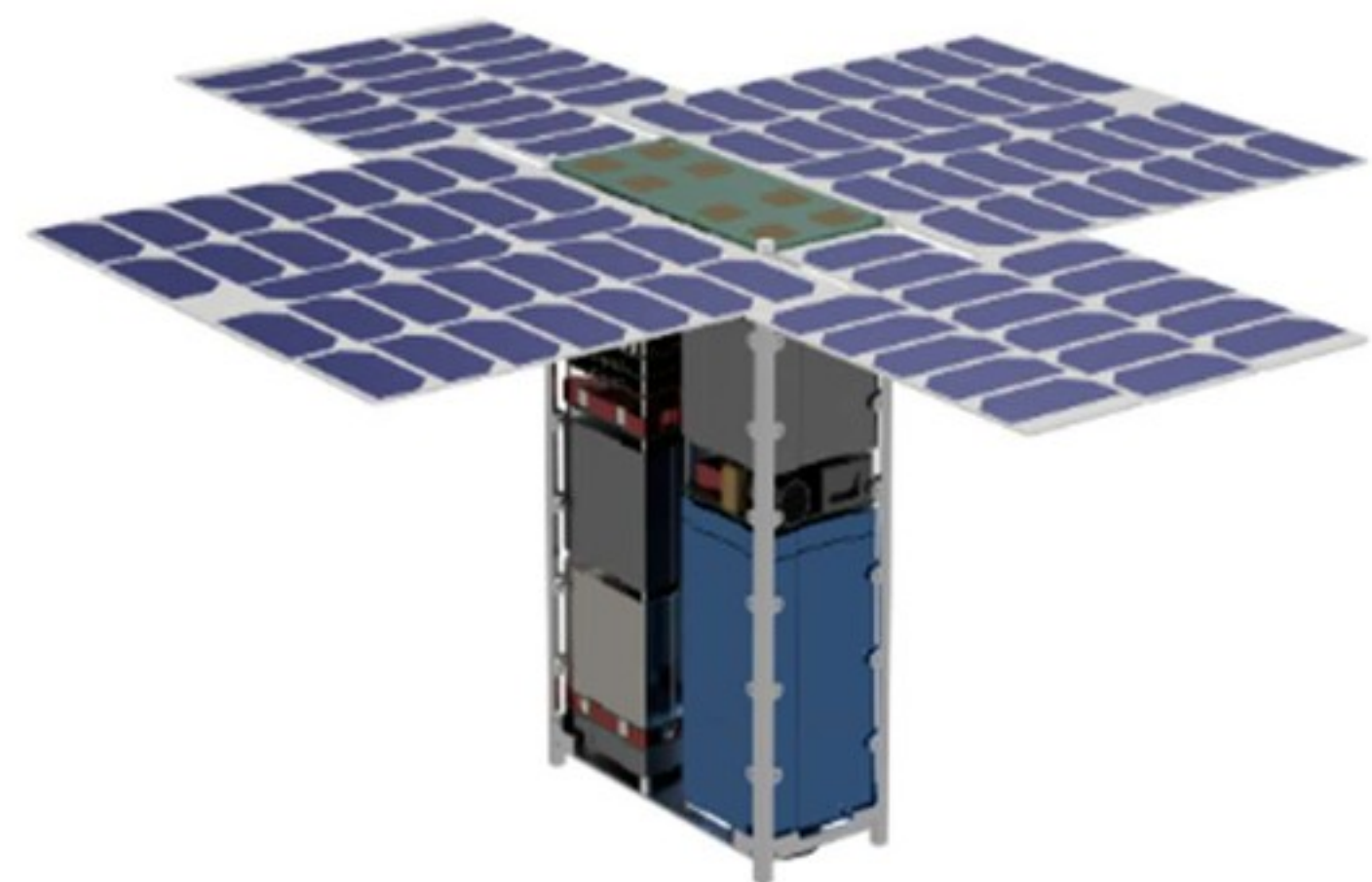


Figure 11: Representation of Proposed Satellite

5.1 On-Board Computer (OBC): The proposed satellite OBC leverages the Xilinx Zynq-7000 FPGA SoC, combining dual-core ARM Cortex-A9 processor with FPGA fabric offering support for high-speed data processing, essential for both Dual-Frequency Synthetic Aperture Radar (DFSAR) and HSI applications. With its capability to handle data rates exceeding 1 Gbps via LVDS interfaces, the SoC provides the necessary throughput for capturing and processing large volumes of high-resolution imaging data in real time.

5.2 Communication System - Data Relay with Chandrayaan -2 or Chandrayaan -4: The data relay between the Constellation and the Chandrayaan satellites is proposed to be in the S-band. With 256-QAM data rates with a 40Mhz bandwidth, the data rate is 320 Mbps, with a link margin of 22 dB.

5.3 Communication System - Inter Satellite Link (ISL) & Direct to Earth (DTE) Beacon: The SDR transceiver supports Ku-band (12-18 GHz) for inter-satellite links (ISL) and L-band for DTE beacon signals. With Ku-band antennas mounted on opposite faces (X+ and X-) and an L-band phased array antenna on the Z+ face. The system operates at a 56 MHz bandwidth (Ku-band) using 256QAM, achieving a data rate of 448 Mbps. The transceiver utilizes 36 dBm transmit power with typical amplifier efficiency and a 4x4 patch antenna with 12 dBi gain. For a 300 km link distance, the path loss is calculated as 121.3 dB. Considering the transmit power, antenna gains, and a receiver sensitivity of -90 dBm, the link margin is approximately 3.5 dB.

5.4 Electrical Power System (EPS): The Proposed EPS consists of 2 radiation shielded PC104 compliant PCB(s) and a 4S4P battery pack.

The Solar panels can produce an instantaneous power of approximately 97 Watts (from 96 cells at 32% Efficiency based on Azur 4G32C Cells). The battery pack used is a 4S2P-designed pack with heaters, integrated charging, and BMS, The cells used are INR21700 M50 with a capacity of 5000 mAh (~18 watt-hours per cell).

Energy generated per orbit	149 Wh
Energy Consumed Per orbit	~95.6 Wh
Energy Margin	53.4 Wh
Battery Storage Capacity	288 Wh
Battery DOD	~60%

Table 4: Approximate Power Budget

5.5 Attitude and Orbit Control System (AOCS): Each CubeSat is equipped with a Scalable ion Electro spray Propulsion System (S-iEPS) manufactured by MIT SPL. Eight MEMS-based ion emitter arrays with thousands of microtips dispersed over an 8 cm² active emission region are part of this system. With less than 1.5 watts of power consumption, the thrusters provide 74 μ N of force at a specific impulse of 1200 seconds, which is enough for orbit maintenance and precise maneuverability. Each thruster can operate for 40–70 h before a decrease in emission current is seen, and it can reach a delta-v of 0.25 m/s over around 9 h. The overall anticipated delta-v requirement for the mission per year is approximately 87.5 m/s. We calculated delt-v using the impulse-momentum relationship

$$\Delta v = \frac{Thrust \cdot Time}{mass}$$

The system can provide delta-v of 233.37 m/s over one year for a 10 kg CubeSat which is sufficient for station-keeping during the mission. (Krejci et al., 2015)

Reaction wheels chosen for the CubeSat are the Nano-avionics 4RW0, occupying a volume of 0.5U. The desaturation of reaction wheels is supported by S-iEPS thrusters.

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